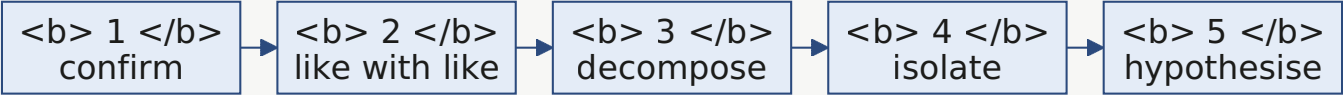


1 The ladder



Crux. You never climb past a rung you have not done, and most wrong answers are somebody starting at rung five because they arrived with a theory.

2 What each rung needs

Rung	Needs	Goes wrong when
Confirm	Both totals, one definition	Delivered meets booked
Like with like	Equal windows, same segments	13 weeks meets 11
Decompose	Customers, orders, revenue	The total is called a finding
Isolate	The same split per segment	A blend hides the segment
Hypothesise	Something outside the data	A cause is stated as fact

3 Grouping by key

```
totals = {}
for order in ORDERS:
    k = order["quarter"]
    totals[k] = totals.get(k, 0) + order["amount"]
```

Count adds one. Revenue adds the amount. Distinct customers uses `setdefault(k, set()).add(...)`.

4 The decomposition has to close

REVENUE = CUSTOMERS x ORDERS PER CUSTOMER x REVENUE PER ORDER

Factor	Q1	Q2	Change
Customers	69	69	flat
Orders per customer	1.65	1.25	down 24.6%
Revenue per order	1.84L	2.17L	up 18.0%
Revenue	2.10cr	1.87cr	down 11.0%

Crux. Multiply the three factor changes and they must land on the revenue change, or a factor is missing and nothing built on the decomposition is safe.

5 return, never print

return Hands the value back, so the next line can use it

print Draws on a screen and hands back `None`

A function that prints cannot be built on. The tell is `TypeError: 'NoneType' object is not subscriptable` one line later.

6 Defaults are decisions

`KeyError: 'discount'` means the key is not there. Before reaching for `.get()`, count how many records and check whether the absence is concentrated.

Choice	What it costs
Drop the row	Revenue falls and the count changes
Default to zero	The count holds, revenue is understated
Keep and flag	Nothing lost, somebody has to look

Crux. There is no free option, so write the reason beside the default, because on Wednesday somebody from Finance asks for it.

7 A finding, not an opinion

A cause with a test beside it is a finding. A cause without one is an opinion with numbers attached.

Retail-Plus down 49 percent against Retail-Core's 5 is a finding. "The reorder button broke" is a hypothesis until reorder events per member either side of the window say so.